BASHA

Muhammad Hawatif Basharuddin Rabbani Tahammal

Software Engineer

bashatahamal.github.io

+6285643755459 — bashatahamal@gmail.com — linkedin.com/in/bashatahamal — Cikarang, Bekasi (17831)

Summary

A versatile Software Engineer with a strong foundation in machine learning, software development, and DevOps. With over 5 years of professional experience, I have successfully led and developed AI-driven applications, including LLM-based conversational assistants, OCR automation, computer vision solutions, and recommendation systems. My expertise spans AI applications, Dev/MLOps, cloud optimization, and AI-driven automation, with a proven track record of enhancing efficiency and reducing operational costs. Passionate about integrating AI solutions into real-world applications, I have contributed to multiple industries, including aquaculture, cybersecurity, e-commerce, and agribusiness. I get things done and thrive in cross-functional teams, leveraging technical expertise, leadership, and innovation to deliver impactful, scalable AI solutions.

Experience

Berning — **January 2026 – July 2026**

Software Engineer, AI Systems (Contract) — *United States, US (Remote)*

AI-powered contract sales organization for life science companies, combining agentic AI with commercial expertise.

Leadflow – AI Lead Generation Platform

- Designed the platform's agentic AI core on an MCP tool layer: each capability (research, scoring, enrichment) built as a reusable agent, callable as an API endpoint, a chat tool, and a workflow step.
- Built an in-app AI agent with dozens of tools that searches, filters, sorts, and acts directly on the lead review canvas (accepting or rejecting leads, merging duplicates, bulk field updates, pulling company research).
- Gave the agent a real control surface: streaming replies showing its thinking and a live tool timeline, a composable skill system, a per-conversation model and reasoning-effort picker, and structured questions so it asks instead of guessing.
- Built a multi-step AI research agent that autonomously gathers and synthesizes company and market information to support lead scoring and outreach.
- Built the core LLM-based lead scoring engine, then redesigned it into a multi-client architecture that scaled from single-client to multi-client with zero breaking changes, on a deterministic rubric so scores stayed traceable.
- Built a duplicate-detection and merge engine for companies and leads, with dry-run previews and transactional apply, wired into the agent as scoped merge tools.
- Rebuilt the internal lead review tool from a single-file prototype into a modular React application: a server-virtualized, spreadsheet-style canvas with filtering, sorting, bulk edits, detail drawers, comments, and live job tracking.
- Built a multi-provider email verification waterfall with automatic fallback and defensive validation, so outreach ran on addresses that actually resolve.
- **Tools Used:** Python, FastAPI, LangGraph, DeepAgents, MCP, PostgreSQL, Redis, React, TypeScript, Docker

HDCE Inc. – TIB Finance — **February 2025 – May 2025**

AI Engineer (Freelance) — *Canada, CA (Remote)*

Canadian IT and cybersecurity firm with a fintech subsidiary (TIB Finance) focused on payment processing and financial automation.

Ran AI research and fast prototyping across multiple initiatives in a concentrated 3-month period. Key initiatives included:

Real-Time Audio & Multilingual Systems

- Built a real time multilingual audio translation system leveraging WebRTC and Meta Seamless model for low latency speech processing.
- Engineered an end to end pipeline for speech-to-text, translation, and text-to-speech including live transcription display and synthesized audio output.

LLM Agents, RAG Systems & Knowledge Automation

- Built LLM agents for documentation, coding assistance, and transforming email content into structured knowledge.
- Designed a multi agent orchestration framework, real time streaming interface, and a vector-based KMS for long-context retrieval.
- Created an MVP developer assistant agent that learns from websites, codebases, and email Q&A to generate guides and summaries.
- Implemented RAG pipelines and delivered internal agents for JSON cleanup, recipe generation, and enhanced chatbot workflows.

Marketing & Social Media Automation Agents

- Developed an AI engine that generates, refines, and schedules social posts, with conversational editing via local LLM.

- Built a firecrawl powered URL ingestion pipeline that converts webpages into post ready content with human-in-the-loop approval.
- Integrated Arcade API publishing, Slack notifications, LangSmith tracing, and delivered a containerized Postgres/Redis backed agent platform.

Financial Automation

- Implemented automated workflows using the Interactive Brokers (IBKR) API for data retrieval and monitoring.
- Integrated existing financial logic into a modular and scalable containerized Python-based automation system.
- **Tools Used:** Python, JavaScript, FastAPI, Node.js, Ollama, LangChain, LangGraph, Firecrawl, Agent Inbox, LangSmith, Postgres, Redis, Arcade API, Slack API, IBKR API, Pandas, Numpy etc.

eFishery **July 2023 – February 2025**
Mid AI Research Engineer *Bandung, ID*
eFishery is Indonesia's first aquaculture technology unicorn and one of the largest companies in the sector globally. It provides innovative solutions for fish and shrimp farming, aimed at enhancing efficiency, productivity, and sustainability in aquaculture.

AI Research Engineer

– LLM Applications

- Initiated and led AI research projects on Large Language Models (LLM), starting with functionality in Retrieval-Augmented Generation (RAG) and real-time database integration. The project later became known as Mas Ahya.
- Created, engineered, and led the breakthrough development of Mas Ahya (a generative-AI-powered conversational assistant for aquaculture). As the principal engineer of Mas Ahya, I maintained and improved the core system, releasing it for internal farmer use. It was presented at Microsoft AI Day by Microsoft CEO Satya Nadella and featured in various news outlets and social media platforms
- Developed internal LLM application, enabling the internal shrimp team to receive better recommendations on which ponds are ready for harvest via conversational interactions.
- Initiated and led the adoption of LLM-based automated testing and evaluation to strengthen engineering processes, particularly in QA testing and performance assessment of LLM applications.
- Inspired the development of an OCR LLM-based solution that reduced and saved engineering costs.
- Planned LLM Apps MultiAgent systems roadmap and conducted proof of concept (PoC) with capabilities integration with WhatsApp chat/non-text media, including voice (text to speech, speech to text), image, and multi-step data confirmation.

– MLOps

- Achieved up to 80% cost reduction in LLM services operations through performance optimization and resource efficiency improvements.
- Led cloud cost analysis, implementing strategies that resulted in up to 60% monthly cost savings through usage optimization.
- Ensured smooth deployment and operational efficiency of AI services.
- Conducted AI service assessments, identifying and addressing areas for cost reduction and performance enhancement.
- Contributed to designing end-to-end Edge Devices and AI roadmap for seamless integration and deployment.

AI Squad Leader

- Squad & Tech Implementation Leadership
  - Led ML/AI development driving cross-functional collaboration with engineers, PMs, and stakeholders.
  - Managed sprint planning, reviews, and retrospectives to ensure efficient Agile execution.
  - Oversaw the implementation and maintenance of ML/AI services across AWS, GCP, Azure.
  - Enforced code review standards and integrated automated quality control tools for reliability and scalability.
  - Ensured seamless ML service deployment across cloud environments.
- Technical Council Contribution
  - Regular speaker to the Tech Council, a bi-weekly multi-division forum involving VP, engineers, and PMs.
  - Critiqued and reviewed technical proposals and implementation strategies to ensure optimal execution.
- Supervised AI Projects
  - Optical Character Recognition (OCR) Solutions
    - Automated sensitive data masking for bank statements (BNI, BRI, BCA, CIMB), processing up to 500-page documents with RabbitMQ for efficient workload distribution and manual error reduction.
    - OCR enhancement with GPT-4.0, cutting processing time, manual data collection, and engineering costs.
    - Document detection and classification for eKYC, utilizing YOLOv8 to optimize ID verification while reducing unnecessary OCR processing costs through manual filtering.
    - Custom OCR system with Azure Document Intelligence, replacing third-party services and cutting costs.
- **Tools Used:** Python, Langchain, Langgraph, Langfuse, FastAPI, AWS, GCP, Azure Platform, Nomad, Kubernetes, Grafana, GPT, LLM, Ollama, Huggingface, Alembic, Pandas, Polar, PyTorch, PostgreSQL, MongoDB etc.

HDCE Inc. **October 2024 – December 2024**
AI Engineer (Freelance) *Canada, CA (Remote)*

Canadian IT and cybersecurity consulting company specializing in system management, network security, cloud services, and technical support for business clients.

AI & Machine Learning Training

- Mentored and guided their team on local LLM model training, fine-tuning, and implementation, providing hands-on support for real-world applications.

Proof of Concept Development

- Designed and implemented a phone call voice-based authentication system, integrating voice recognition, biometric security, speech-to-text, text-to-speech and Twilio's API for secure and scalable deployment.
- **Tools Used:** Python, PyTorch, CUDA, SpeechBrain, Whisper, DeepSpeed, Huggingface, Scikit-learn, Twilio, Flask

Great Giant Foods **July 2024 – September 2024**
AI Engineer (Freelance) *Yogyakarta, ID*
Great Giant Foods (GGF) is a leading agribusiness company in Indonesia, specializing in the production and export of high-quality fresh and processed fruits, juices, meats, and dairy products.

OCR Based Solution

- Developed OCR-based desktop applications to enhance the efficiency of banana harvest form record-keeping using GPT-4o, ensuring easy maintenance by optimizing prompt engineering assets.
- Developed API-based OCR Services leveraging GPT-4o for seamless integration and communication with other services.

LLM Based Application

- Built and Deployed End-to-End RAG-based (Retrieval Augmented Generation) applications using GPT-4o with a Streamlit web application to provide internal educational resources on banana cultivation, backed by grounded data knowledge.
- **Tools Used:** Python, FastAPI, Langchain, OpenAI, Streamlit, Pandas, PyQt5, Docker

Yuna & Co. **April 2021 – June 2023**
Machine Learning & Software Engineer, DevOps *Jakarta, ID*
A women's fashion and lifestyle e-commerce platform & personal styling based on Artificial Intelligence.

Machine Learning related responsibility

- Developed a user clustering service for enhanced data analysis.
- Created a product recommendation service based on similar products, with a caching strategy for efficient output.
- Implemented image understanding functionalities, such as object detection for duplicate image search, style card generator functionality (outfit reference images containing recommended products), and color picker based on similar RGB values.
- Developed a service for product searches based on similar characteristics using images and/or tagging. This was utilized for functionalities such as user wardrobe (personal collection and recommended outfits), style card generator (admin tool for faster generation of style cards), and similar product suggestions for the main website shop page.
- Implemented collaborative filtering to recommend products based on user feedback and constructed generic user profile characteristics.
- **Tools Used:** Python, Scikit-learn, OpenCV, Google Vision, Numpy, Pandas and other related libraries/frameworks.

Software Engineer related responsibility

- Built and maintained machine learning, image, and data API services using Python FastAPI.
- Developed and managed task scheduler services using Celery, Django, MongoDB, and RabbitMQ.
- Responsible for building and maintaining Python API services and contributing to NodeJS API maintenance.
- Designed, created, and maintained asset management services with image manipulation capabilities (padding, cropping, resizing, color changing) and cached derivative images for faster load times.
- Designed, maintained and implemented a message broker for communication between services using RabbitMQ.
- Contributed to debugging and migrating the NodeJS API application to a new Typescript framework.
- Added listeners and notifiers to the PostgreSQL database for data validation.
- Managed data migration, cleansing and provided data summaries for further analysis.
- **Tools Used:** Python, OpenCV, Flask, FastAPI, Django, Celery, Pandas, RabbitMQ, Postgres, MongoDB, NodeJS, TypeORM, Typescript, Cloudinary etc.

DevOps related responsibility

- Managed and handled all aspects of development operations, including deployment, server management, application containerization, automation and etc.
- Initiated and implemented the use of Kubernetes for application deployment along with CI/CD pipeline deployment using Jenkins, Ansible, and related frameworks.
- Dockerized and containerized all Yuna applications for microservices deployment into GCP and AWS.

- Created, designed, tested, and maintained infrastructure for all services and microservices-based applications using Kubernetes.
- Handled network traffic using Ingress, Nginx, load balancers, and Cloudflare.
- Built automation tasks such as Selenium testing, Telegram Bot - Jenkins integration (sending Jenkins commands via button in Telegram chat group), updating image product sets on Google Vision, automatic database backups, and WhatsApp server automation.
- **Tools Used:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Kubernetes, Ansible, Terraform, Kubernetes Operations (KOps), Jenkins, Cloudflare, Telegram bot UI, Bitbucket webhook, Selenium, PyWinAuto, Bash Script etc.

Freelance Project — **September 2020 – February 2021**
Professional Assignments (ML Researcher) — *Yogyakarta, ID*
Traffic monitoring system for counting vehicles and detecting traffic violation based on video feed

Intelligence Traffic Monitoring System

- Led the project, providing strategic solutions and innovative ideas.
- Prepared, trained, and fine-tuned the object recognition model using a custom low-resolution dataset.
- Developed a web-based system interface.
- Implemented algorithms for traffic violation detection and vehicle counting through video processing (object recognition & object tracking).
- **Tools Used:** Python, Tensorflow, PyTorch, OpenCV, Flask, Javascript, HTML/CSS

Freelance Project — **July 2020**
Professional Assignments (ML Researcher) — *Malaysia, MY*
Detect and recognize road damage on images pixel by pixel and put mark/bounding box on it.

Road Damage Detection and Recognition

- Implemented methods to identify and classify road damage types using Yolov4 and Yolov5 Convolutional Neural Network architectures.
- Trained and fine-tuned architecture models with the prepared dataset.
- Compared the latest methods for each required output type using LinkNet, UNet, and Feature Pyramid Network (FPN) with varied Convolutional Neural Network backbones.
- Compiled final reports for testing and model evaluation.
- **Tools Used:** Python, Tensorflow, PyTorch, OpenCV, Google Colaboratory

Education

Gadjah Mada University (UGM) — **August 2016 - November 2020**
Bachelor of Electronics and Instrumentation — *GPA 3.61 out of 4.00*
Minors: Computer Vision; Machine Learning

Undergraduate thesis project

- **Topic:** Quran Tajweed Rules Recognition
- Conducted research to develop methods for recognizing Tajweed rules using image-based input (document processing).
- The end-to-end project involved designing the method and its evaluation scheme, building a dataset of Arabic characters, implementing image processing techniques consisting of detecting the occurrence of reading law characters using template matching, developing a segmentation algorithm for Arabic words based on pixel values, and utilizing Convolutional Neural Networks for character recognition. Additionally, a web-based user interface was developed.
- **Tools Used:** Python, Tensorflow, OpenCV, Flask, PySimpleGUI, Numpy, Pandas, Javascript, HTML/CSS

Skills

AI/ML LLMs (RAG, Langchain, Langgraph, MCP), Computer Vision (OCR, Object Detection, Image Processing), NLP, Speech Recognition, Recommendation Systems
Backend Python (FastAPI, Flask, Django), TypeScript (Node.js, React), API Development, Microservices, Event-Driven Systems
Databases PostgreSQL, MongoDB, VectorDB
Messaging RabbitMQ, Celery
DevOps AWS, GCP, Azure, Docker, Kubernetes, Nomad, Terraform, Ansible, Jenkins, Bitbucket Pipelines, Git
Automation Celery, Bash, Chatbot Integration (WhatsApp, Telegram), OCR Services, LLM evaluation
Common Libs PyTorch, TensorFlow, Huggingface, OpenAI, Pandas, NumPy, OpenCV, Langfuse, LangSmith, Ollama, Streamlit etc.
Cloud Platform Azure Platform, Google Cloud Platform, Amazon Web Services
Soft Skills Fast learner, Leadership, Agile, Collaboration, Technical principal, Strong problem-solving skills
Languages English, Bahasa Indonesia